Lab Report – EE-170L Computer Programming (Lab 10)

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Objectives

- *Understand **function call by value**.*
- *Understand **function call by reference**.*
- *Learn about **default parameter values** in functions.*
- *Apply these concepts in practical C++ programs.*

Theory

***Call by Value**: A copy of the variable is passed to the function. Changes inside the function do not affect the original variable.*

***Call by Reference**: The actual memory address is passed. Changes inside the function directly affect the original variable.*

***Default Parameters**: Functions can have default values for parameters, which are used if no argument is provided during the call.*

Lab Tasks

Task 1: Function with Parameters

```
#include <iostream>                 // include input/output library

using namespace std;             // use standard namespace


void multiply(int num1, int num2) {  // function with two integer parameters

   int product = num1 * num2;        // calculate product of num1 and num2

   cout << "Product = " << product;   // display result inside function

}
```

```
int main() {                    // main function starts
   multiply(6, 7);              // call multiply function with two numbers
   return 0;                    // end program
}
```

Task 2: Function with Return Value

```
#include <iostream>                // include input/output library
using namespace std;               // use standard namespace

int getSquare(int number) {            // function with one integer parameter
   int square = number * number;       // calculate square of the number
   return square;                      // return the result
}

int main() {                              // main function starts
   int result = getSquare(9);             // call function and store returned value
   cout << "Square = " << result;         // display the square
   return 0;                              // end program
}
```

Results

- *Task 1 displayed:* ***Product = 42*** *(for inputs 6 and 7).*
- *Task 2 displayed:* ***Square = 81*** *(for input 9).*

Conclusion

In this lab, I learned:

- *How to pass parameters to functions.*
- *How to return values from functions.*
- *The difference between call by value and call by reference.*
- *How default parameters simplify function calls.*

*Functions make programs **modular, reusable, and efficient**.*